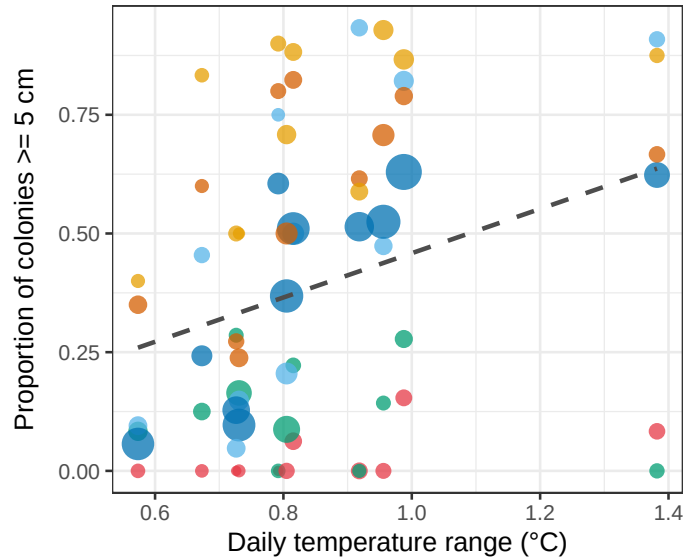


Three mechanisms linking environment to Pocillopora size–frequency distributions

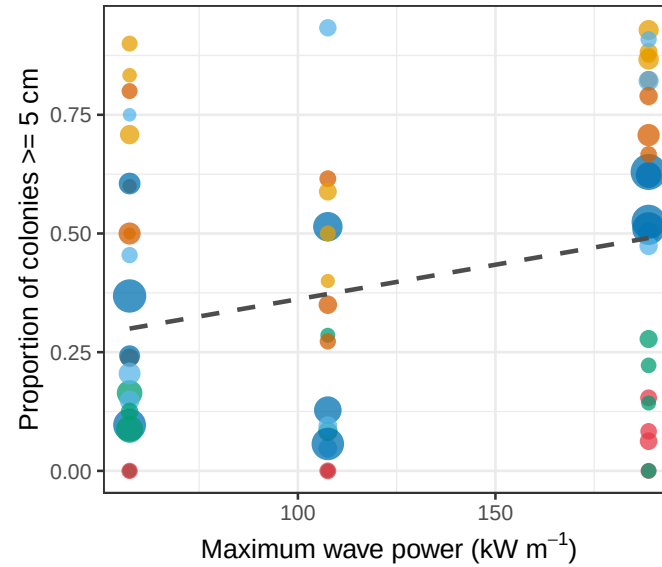
A Thermal variability

Greater daily range → adult-dominated size structure



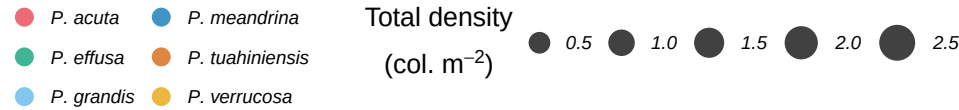
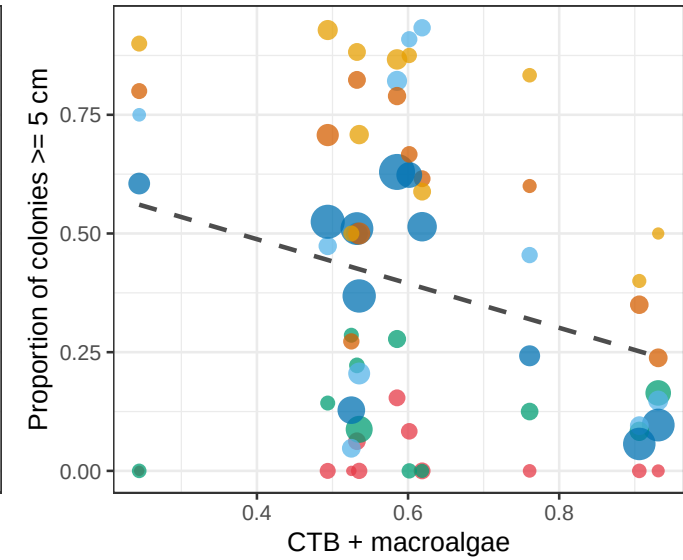
B Wave exposure

Higher wave power → size-selective dislodgement of small colonies



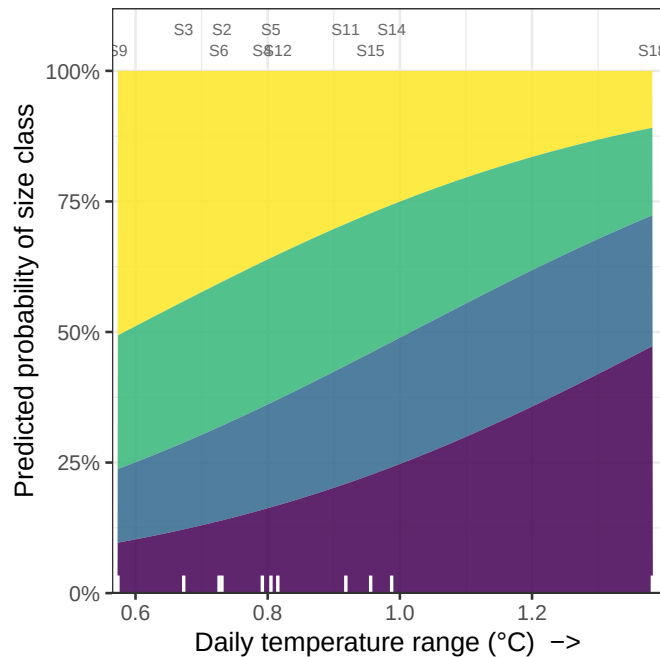
C Hard substrate availability

Higher CTB + macro cover → smaller colonies (significant in *P. meandrina*, *P. tuahiniensis*)



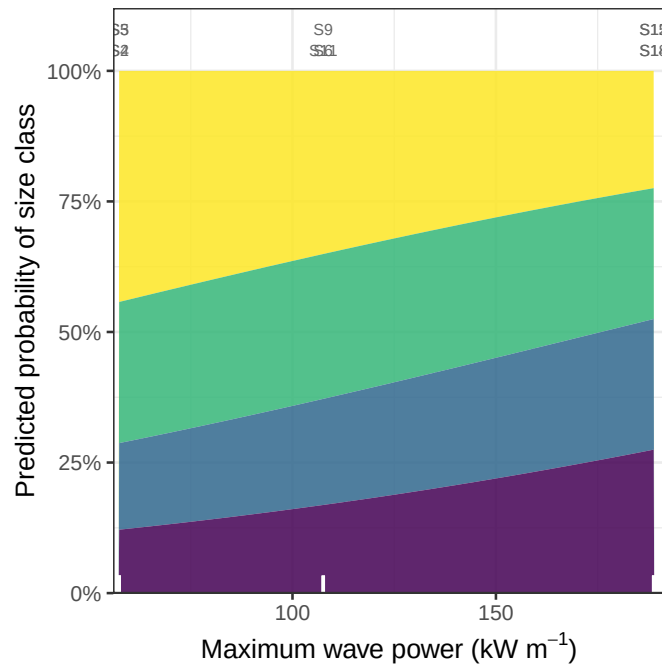
D Predicted size structure – thermal variability

All Pocillopora spp. pooled | OLR (1|Species) + (1|site) | Juvenile <– → Adult



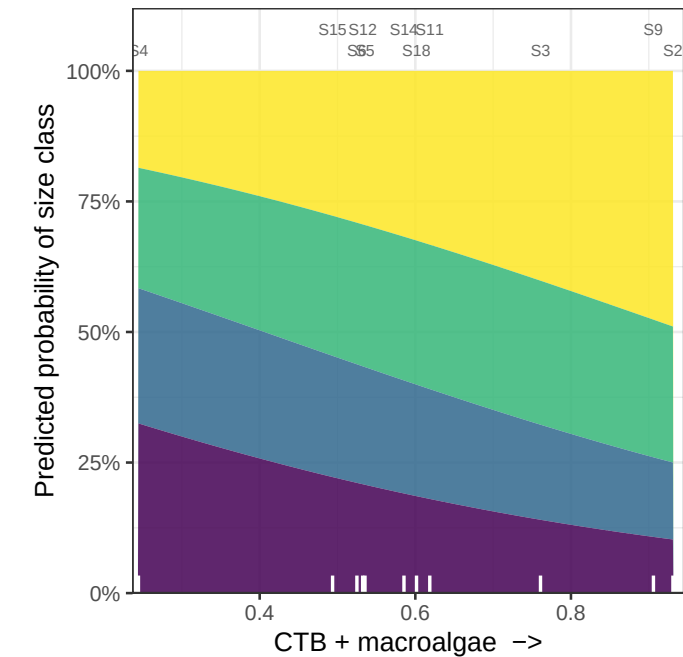
E Predicted size structure – wave exposure

All Pocillopora spp. pooled | OLR (1|Species) + (1|site) | Juvenile <– → Adult



F Predicted size structure – hard substrate

All Pocillopora spp. pooled | OLR (1|Species) + (1|site) | Juvenile <– → Adult



Top: site × species combinations; size ... total colony density; dashed line = cross-species OLS trend.
 Bottom: OLR-predicted size class probabilities (all spp. pooled; species + site random effects); rug ticks = site locations. n = 11 sites.